

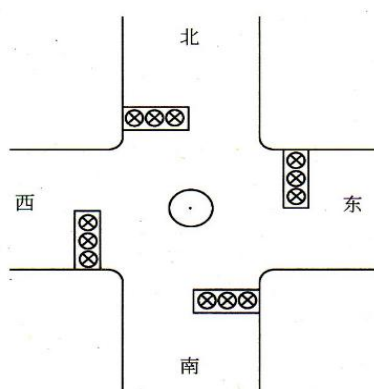
交通灯实验

1、实验目的:

使用 HX4S20 开发板上接流水灯和交通的扩展板，设计一个简单的交通灯，东西路口和南北路口分别由红、黄、绿控制，了解交通灯的亮灭规律。

2、实验原理:

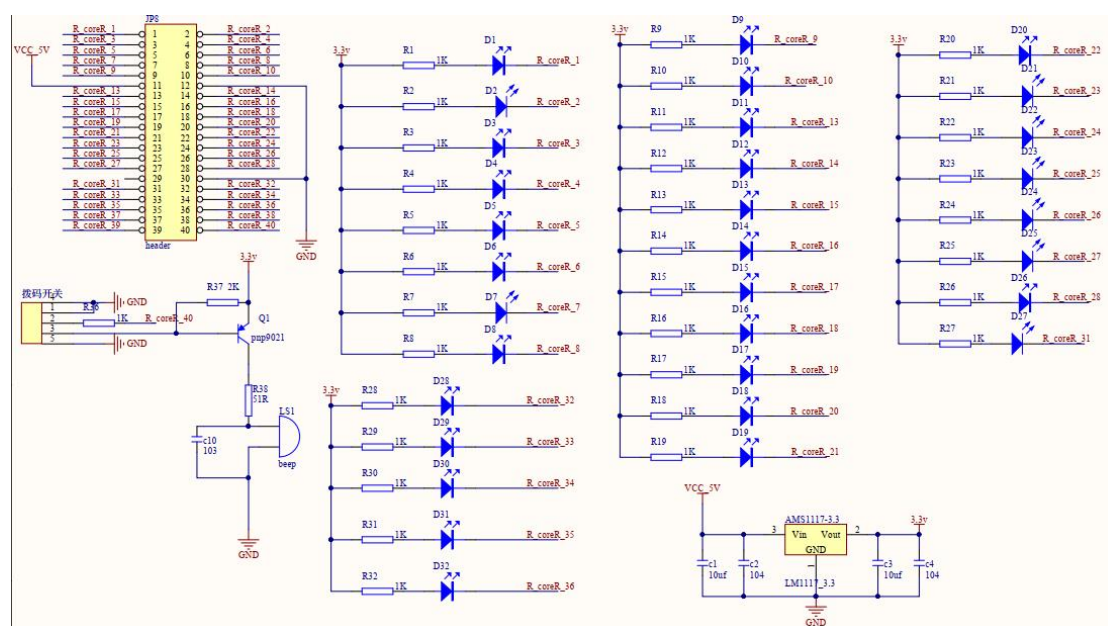
交通灯的显示有很多方式，如十字路口、丁字路口等，而对于同一个路口又有很多不同的显示要求，比如十字路口，车子如果只要东西和南北方向通行就很简单，而如果车子可以左右转弯的通行就比较复杂，本实验仅针对最简单的南北和东西直行的情况。根据以上情况，设计交通灯控制器，从整个方向红灯切换到绿灯需要 25S，其中绿灯转换成黄灯需 20S，其中黄灯占 5S，到绿灯共计 25S，根据这思路如下图所示：



交通灯控制要求

东西	信号	绿灯	黄灯	红灯	
	时间	20S	5S	25S	
南北	信号	红灯		绿灯	黄灯
	时间	25S		20S	5S

根据开发板的原理图，可得到以下资料：



根据硬件原理图所示，发光二极管，所有的阳极都接通 3.3V 的正电压，也即高电平，所以如果我们想要发光二极管导通的话，需要在阴极接通低电平，就可以让 LED 亮起来。

3、实验程序：

```
module traffic(clk, r1, y1, g1, r2, y2, g2);
    input    clk;
    output reg [1:0]  r1,y1,g1;
    output reg [1:0]  r2,y2,g2;
    parameter yellow_time = 5;
    parameter green_time = 20;
    reg [1:0]  state;
    reg [4:0]  count;
    always @(posedge clk)

        case (state)
            0 :
                begin
                    r1 <= 2'b11;
                    y1 <= 2'b11;
                    g1 <= 2'b00;
                    r2 <= 2'b00;
```

```
        y2 <= 2'b11;
        g2 <= 2'b11;
    end
1 :
    begin
        r1 <= 2'b11;
        y1 <= 2'b00;
        g1 <= 2'b11;
        r2 <= 2'b00;
        y2 <= 2'b11;
        g2 <= 2'b11;
    end
2 :
    begin
        r1 <= 2'b00;
        y1 <= 2'b11;
        g1 <= 2'b11;
        r2 <= 2'b11;
        y2 <= 2'b11;
        g2 <= 2'b00;
    end
default :
    begin
        r1 <= 2'b00;
        y1 <= 2'b11;
        g1 <= 2'b11;
        r2 <= 2'b11;
        y2 <= 2'b00;
        g2 <= 2'b11;
    end
endcase
always @(posedge clk)
begin
```

```
        if (count == 0)
        begin
            if (state == 3)
                state <= 0;
            else
                state <= state + 1;
        case (state)
            2'b00 :
                count <= yellow_time;
            2'b01 :
                count <= green_time;
            2'b10 :
                count <= yellow_time;
            default :
                count <= green_time;
        endcase
    end
    else
        count = count - 1;
    end
endmodule

module clkdiv(clkin,clk1Hz); //分频器
input clkin; //输入时钟频率 50MHz
output clk1Hz; //1Hz
reg clk1Hz;
reg [24:0]cnt1;

always@(posedge clkin)
    if(cnt1==24999999) //50000000 分频
        begin
            clk1Hz<=~clk1Hz;
            cnt1<=0;
        end
    else cnt1<=cnt1+1;

endmodule
```

4 、实验仿真:

根据描述语言进行仿真测试，分析波形。

5 、管脚约束:

```
set_pin_assignment { EG } { LOCATION = K16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { ER } { LOCATION = M16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { EY } { LOCATION = H16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { NG } { LOCATION = N14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { NR } { LOCATION = P16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { NY } { LOCATION = N16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { SG } { LOCATION = G14; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { SR } { LOCATION = K15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { SY } { LOCATION = G16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { WG } { LOCATION = D16; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { WR } { LOCATION = E15; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { WY } { LOCATION = M10; IOSTANDARD = LVCMOS33; DRIVESTRENGTH  
= 8; PULLTYPE = NONE; }  
set_pin_assignment { clk } { LOCATION = R7; IOSTANDARD = LVCMOS33; PULLTYPE =  
PULLUP; }
```

6 、下载测试

将流水灯模块接插在开发板 J1 口上。

下载程序之后。可以看到交通灯红绿黄灯切换。

